

# Carcinoma of Breast

- Name:
- Age:
- Sex:
- Occupation:
- Address:

DOA:

DOE:

## Chief Complaints:

1. Swelling – in breast / axilla (lymph nodes)
2. Pain
3. Retraction of nipple
4. Discharge from nipple
5. Wound/ Trauma / Ulceration / Fungation
6. Metastasis symptoms

## History of Presenting illness:

### **1. Lump in Breast:**

- Duration
- Onset
- Rate of growth

### **2. Pain:**

- Duration
- Type
- Radiation
- Aggravating/Relieving factors
- Associated symptoms (fever in mastitis)
- Relation to menstruation (cyclical mastalgia)

### **3. Asymmetry of Breast:**

### **4. Discharge from nipple:**

- Duration
- Quality-nature, colour, odour
- Quantity

### **5. Nipple changes:**

- Retraction(depression)
- Deviation
- Destruction
- Displacement
- Discolouration
- Duplication

### **6. History of trauma** (hematoma, traumatic fat necrosis)

### **7. History of swelling elsewhere** (limb)

### **8. History of fever** (mastitis)

### **9. History of loss of weight/appetite**

### **10. History of related to metastasis**

- History of bone pain low back pain, altered sensation like sense of position and vibration, lower limb weakness, features of paraplegia,
- History of jaundice, Abdominal distention
- History of chest pain/ breathlessness
- History of cough with hemoptysis
- History of convulsions, loss of consciousness, vomiting, limb weakness, headache, visual disturbances, behavioural changes (psychological changes)
- History of haematuria

### **11. History of Radiation exposure (15-20years ago)**

### **12. History of Menstrual irregularities , Abdominal pain**

#### **Past History:**

- History of contraception
- History of similar complaints in past (fibroadenosis, TB)
- History of colorectal or pancreatic carcinoma
- History of Radiation exposure
- History of surgery (radial incision)
- History of nipple Retraction ( in abscess)
- History of DM, HTN, TB, EPILEPSY, ASTHMA

#### **Personal history:**

##### **➤ Menstrual History**

- Age at menarche
- Cycles (regular/irregular)
- Duration
- Cyclical mastalgia
- Age at menopause

##### **➤ Obstetric History:**

- Marital status
- Number of pregnancies
- Still birth , Abortions, Spacing
- Age at first childbirth
- Breastfeeding,
- Last child birth
- Use of OCPs

##### **➤ Appetite**

##### **➤ Diet**

##### **➤ Bowel and Bladder:**

##### **➤ Sleep**

##### **➤ Habits**

**Family History:**

- History of similar complaints in 1<sup>0</sup> relatives ( common in sibling than mother)
- History of Ovarian Ca, Gastric Ca & Pancreatic Ca

**Drug History:**      HRT  
                              OCPs

**Allergy History:****General Physical Examination:**

Here is a \_\_\_\_ year old \_\_\_\_ patient, \_\_\_\_\_ built and nourished and \_\_\_\_ co-operative.

Pallor, Icterus, Cyanosis, Clubbing, Koilonychia, Lymphadenopathy, Edema

**Vitals:**

- Pulse:      (rate, rhythm, character, volume, condition of vessel wall  
                  radio-radial/radio-femoral delay)
- BP: (position and limb in which BP is measured)
- RR:
- Temperature:

**LOCAL EXAMINATION:**

- Position of Patient:
  - Sitting arms by side of patient
  - Sitting arms over the head (Auchincloss method)
  - Bending forward (Fixity to chest is observed)
- Exposure:
- Position of Examiner wrt. Patient:
- Light source:

**Inspection:** ( Inspect in all 3 Positions mentioned)**Comparative inspection of both breasts.**

- Size:
- Shape:
- Position / Extent on both sides: ( 2<sup>nd</sup> – 6<sup>th</sup> ICS)
- Symmetry
- Visible lump:   Quadrant, size, shape
- Level of nipples

**Inspection of affected Breast:**

- Size and Shape of Breast
- Symmetry and Position (pulled towards/ pushed away from mass)
- Nipple
  - Position (drawn up / pushed down)
  - Symmetry (compare to normal side)
  - Prominent / Flattened

- Size & Shape
- Surface
- Displacement ( towards / away from lump)
- Retraction ( circumferential / slit like)
- Discharge
- Ulcer over nipple
- Areola
  - Colour
  - Size changes
  - Cracks / fissures / Ulcer
  - Eczema
  - Discharge
- Skin over the Breast
  - Scars
  - Engorged veins
  - Pigmentation
  - Shininess
  - Redness
  - Skin changes
    - Dimpling (Increases on bending)
    - Puckering
    - Peau d'orange
    - Retraction ( Schirrous Ca Breast)
    - Fungation / Ulceration
- Lump
  - Position: (Quadrant)
  - Number, Size & Shape
  - Extent
  - Surface & Edges
  - Pulsations
  - Skin over swelling
- Ulcer
  - Position / Site
  - Number, Size & Shape
  - Extent
  - Margins Edges & Floor
  - Discharge & surrounding tissue

Edema of Arms:

Inspection of Axilla and Supraclavicular fossa for nodes:

### **Palpation:**

( palpate by palmar surface of fingers quadrant wise including axillary tail of spence and behind nipple)

**Examination of Normal Breast:** palpate all 4 quadrants (firm, lobulated, nodular)

**Examination of affected Breast:**

- Temperature & Tenderness

- Lump
  - Number
  - Position and extent (Quadrant)
  - Size and Shape
  - Surface, Margins
  - Consistency
  - Mobility
  - Fixity to
    - Skin
    - Breast tissue
    - Chest wall
    - Pectoralis Major
    - Serratus Anterior
  - Fluctuation (if cystic in consistency)
  - Trans-illumination test ( If Fluctuation +)
- Nipple
  - Retraction
  - Retro nipple mass
  - Discharge Expression
- Ulcer:
  - Tenderness
  - Size, Depth
  - Edge, Margins, Base
  - Bleeding /Discharge
  - Relation to deeper structure
  - Mobility
  - Surrounding skin

**Examination of Lymph nodes of both sides:** (Axillary, Supra/Infraclavicular)  
**Examination of arm:**

### **Systemic Examination:**

**CVS:**

**RS:** altered breath sounds, features of consolidation or pleural effusion

**PA:** palpable nodular liver, Krukenberg tumours in ovaries in menstruating age group, and ascites

**PR / PV examination:**

**CNS:** neurological deficits

### **Differential Diagnosis:**

#### ▪ **Carcinoma of Breast Differentials**

- |                            |                            |
|----------------------------|----------------------------|
| 1. Fibroadenosis.          | 6. Mastitis.               |
| 2. Traumatic fat necrosis  | 7. Antiboma.               |
| 3. Tuberculosis of breast. | 8. Galactocoele            |
| 4. Bloodgood cyst          | 9. Mondor's disease        |
| 5. Filariasis breast.      | 10. Cystosarcoma phyllodes |

### **Cystic swellings**

- Bloodgood cyst
- Breast abscess
- Hydatid cyst
- Galactocele
- Serocystic disease of Brodie
- Cystic necrosis in carcinoma breast
- Lymph cyst
- Haematoma in breast

### **Massive enlargement**

- Benign hypertrophy usually bilateral
- Giant fibroadenoma (> 5 cm)
- Serocystic disease of Brodie
- Sarcoma
- Carcinoma often when extensively involved
- Filariasis of breast.

### **Hard swellings**

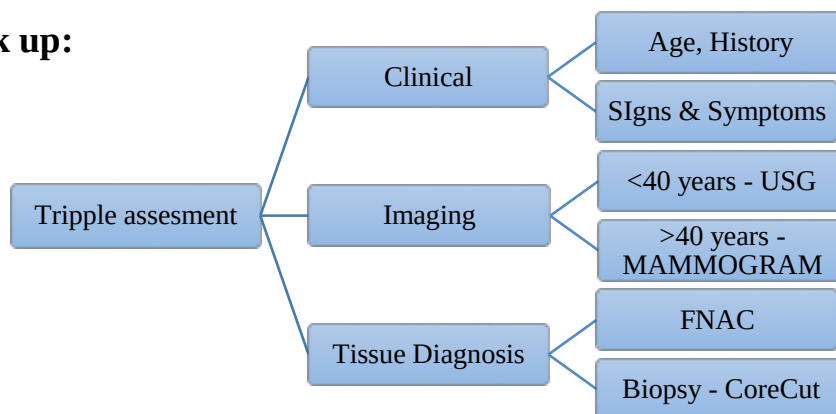
- Carcinoma breast
- Antiboma breast
- Traumatic fat necrosis
- Calcified haematoma
- Fibroadenoma—hard variety

## **DIAGNOSIS:**

Complete diagnosis with side and staging should be given/written in case sheet.  
TNM staging is used.

## **MANAGEMENT:**

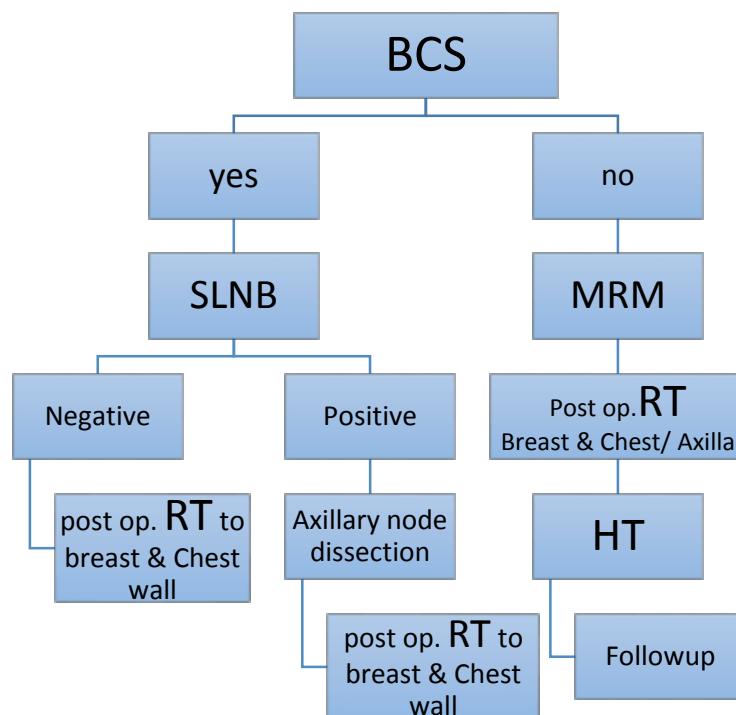
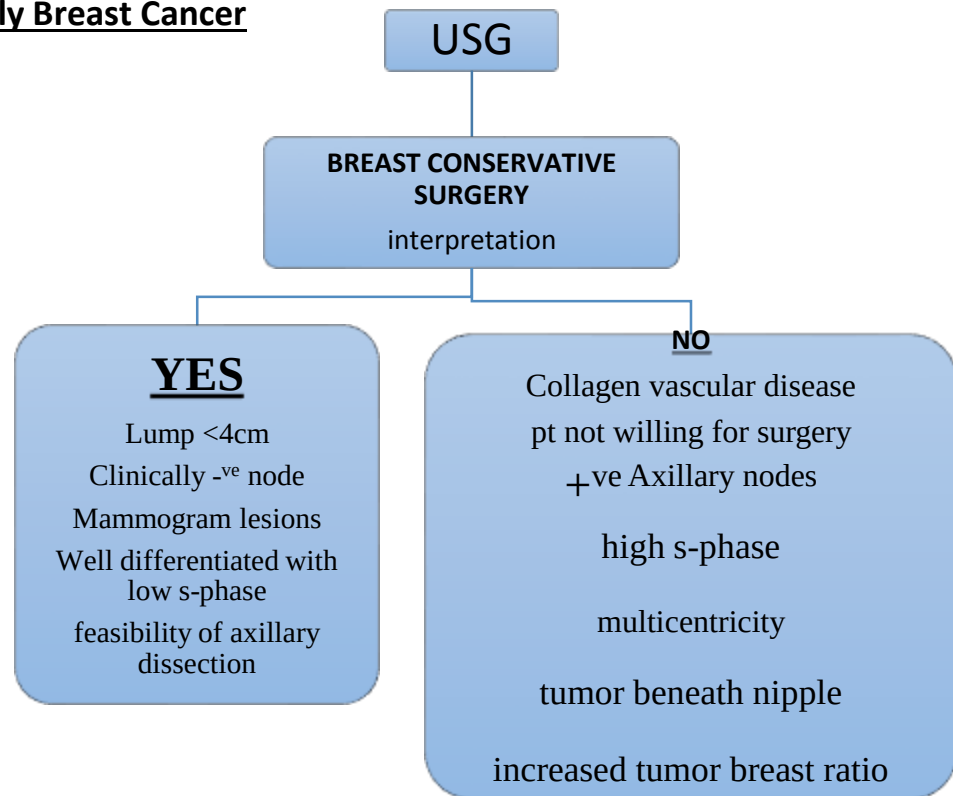
### **Diagnostic Work up:**

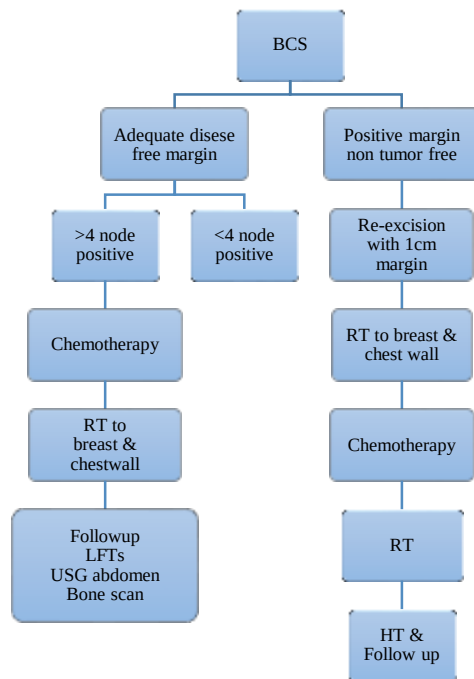


### **STAGING WORK UP:**

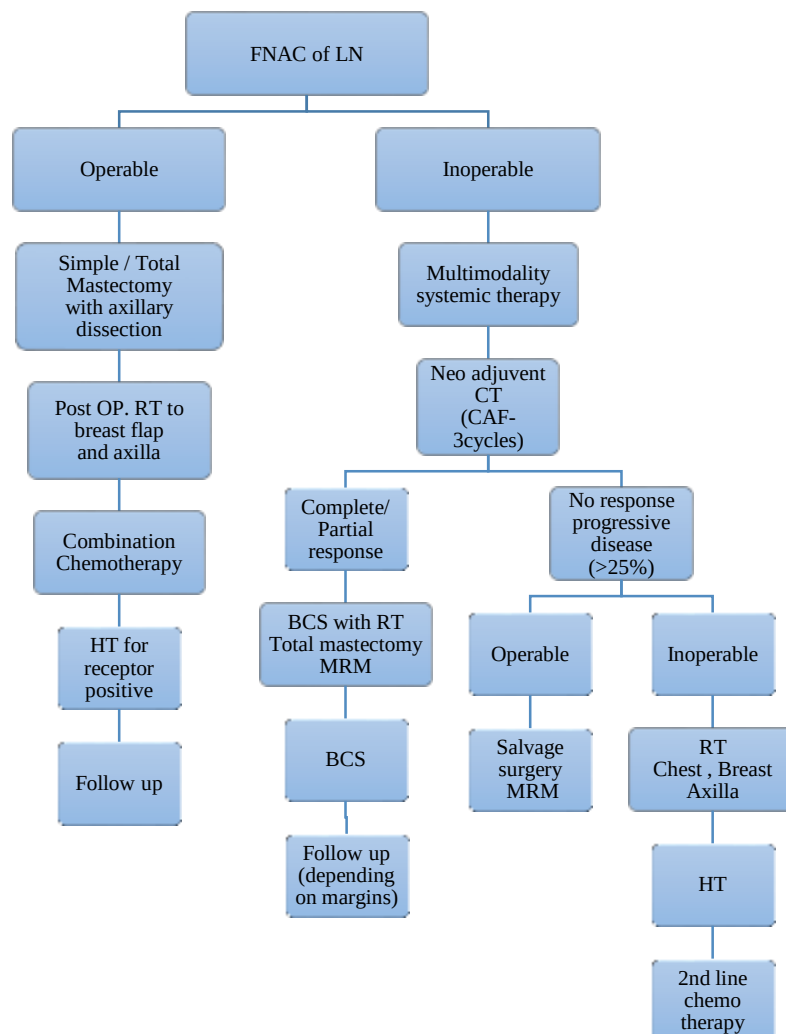
- Clinical Examination
- Mammogram on both sides
- USG abdomen
- Chest X-Ray
- Liver Function Tests
- Bone Scan
- Histo pathology
  - ✓ ER & PR status
  - ✓ HER2 status
  - ✓ Lymph node status

## Management of Early Breast Cancer





## MANAGEMENT OF LOCALLY ADVANCED CA. BREAST



## THYROID

**Name:**

**Age:**

<20 – 40 years

- SNT
- Colloid Goitre
- Pappillary carcinoma thyroid
- Primary thyrotoxicosis

>40 years

MNG  
Hashimotos  
Follicular Carcinoma  
Anaplastic

**Address:** Chikkamagalur and coorg **IDG** endemic

**Sex:** Thyroid disease **8** times & Ca 3 times more common in females

**Occupation:**

### **Chief Complaints:**

Swelling

Pain (Haemorrhage, Infiltration of nerves, Thyroiditis, Anaplastic Ca.)

Alteration in voice

Bulging of eyes

Tremors / Involuntary movements

Palpitation

Increased appetite with weight loss

Diarrhoea ( Medullary Ca.)

Generalised weakness

Easy fatiguability

Generalised increase in sweating

### History of Presenting Illness

**Pain:** ( SOCRATES) ( Whether Pain appeared first or swelling)

Dull aching from start of swelling : Thyroiditis

Sudden onset pricking type of pain that appeared later: Haemorrhage

Dull aching, long standing , appeared later: Malignancy

### **Whether Swelling or Thyroid symptoms appeared first?**

Symptoms precedes swelling : Graves disease

Swelling precedes Symptoms: Plummer's Disease

### **Pressure Symptoms: ( due to Infiltration )**

Difficulty in breathing and swallowing ( on trachea & oesophagus)

Altered voice (on RLN)

h/s/o Horner's syndrome (on cervical sympathetic chain)

Stridor, Haemoptysis (Laryngotracheal complex)

Berry's sign (on carotid sheath)

Syncope (on Carotid body)

Cranial nerve and Brachial plexus symptoms

## Thyrotoxicosis

|                                                      |                                                      |
|------------------------------------------------------|------------------------------------------------------|
| Increased appetite (1 <sup>0</sup> &2 <sup>0</sup> ) | Weight loss (1 <sup>0</sup> &2 <sup>0</sup> )        |
| Heat loss                                            | Heat intolerance (1 <sup>0</sup> &2 <sup>0</sup> )   |
| Amenorrhea                                           | Exertional Breathlessness                            |
| Nervousness (1 <sup>0</sup> )                        | Irritability & Behavioural change (1 <sup>0</sup> )  |
| Insomnia (1 <sup>0</sup> )                           | Tremors (1 <sup>0</sup> )                            |
| muscle weakness (1 <sup>0</sup> )                    | muscle wasting                                       |
| Cold tolerance                                       | Increased sweating (1 <sup>0</sup> &2 <sup>0</sup> ) |
| Diarrhoea (1 <sup>0</sup> &2 <sup>0</sup> )          | Difficulty in closing eyes                           |
| Eyes: (1 <sup>0</sup> )                              |                                                      |

- Bulging – duration, progression
- Redness
- Watering
- Double vision
- Loss of vision

h/o anti-Thyroid drugs

h/o **secondary** thyrotoxicosis

|             |                     |
|-------------|---------------------|
| Palpitation | Ectopic beats       |
| Arrhythmias | Exertional dyspnoea |
| Chest pain  | Ankle edema         |
| CCF         |                     |

## Hypothyroidism

|                                                     |                             |
|-----------------------------------------------------|-----------------------------|
| Weakness                                            | poor appetite > Weight gain |
| Obesity (back & shoulder)                           | Fatigue & lethargy          |
| Swelling- face n whole body                         | Cold intolerance            |
| Loss of scalp hair                                  | constipation                |
| Hoarseness of voice                                 | Dry inelastic skin          |
| Macroglossia                                        | Patchy pigmentation of skin |
| Mask like face(dull expression)                     | Failing memory              |
| Menorrhagia followed by amenorrhea ( due to anemia) |                             |

## Symptoms of Retro sternal Extension

|                                   |             |
|-----------------------------------|-------------|
| Nocturnal dyspnoea                | Dysphagia   |
| Postural cough on lying down      | Haemoptysis |
| Dilated veins over chest and neck |             |

## History of Metastasis

Symptoms suggestive of Hornors syndrome  
Bony pain / swelling  
Chest- cough dyspnoea haemoptysis  
Brain- seizures vomiting  
Significant weight loss & loss of appetite

- **Past History:**

- Exposure to radiation ( papillary carcinoma)
  - Ionising ( R<sub>x</sub> of Tinea capitis, Adenoids, Hodgkins, Hemangioma etc. )
  - Radio Iodine
- H/o previous thyroid swelling (Thyroglossal Cyst)
- H/o Malignancy
- H/o Thyroid surgeries in past ( thyroglossal fistula in TGC S<sub>x</sub>)
- h/o DM, HTN, IHD, Asthma,

**Personal History:**

- Appetite: ( increased in thyrotoxicosis & vice versa)
- Diet:
  - Consuming Brassica family vegetables
  - Water rich in fluorine
  - Type of salt
  - Sea fish deficient of iodine
- Bowel & Bladder: Constipation in Hypo & Diarrhoea in Hyper )
- Sleep: insomnia in Hyperthyroidism
- Habits

**Menstrual History**

- Age at menarche
- Cycles
- Duration
- Flow
- Clots

Oligomenorrhea in Hyperthyroidism

Menorrhagea in Hypothyroidism

**Family History**

- H/o any Thyroid disease in family
- Dysharmonogenesis, Medullary Ca. Thyroid, Primary thyrotoxicosis
- H/o infertility in family ( altered thyroid function)

**Drug History:**

- Eroxin
- Antithyroid drugs
- Beta blockers
- Sulfonyl ureas
- Methyl DOPA
- T<sub>3</sub> T<sub>4</sub> TSH Followup

**Allergy History:**

## **GENERAL PHYSICAL EXAMINATION**

### **Built & Nourishment**

Hyperthyroidism: lean & anxious

Hypothyroidism: Obese, sluggish & Over weight

Malignancy : Cachexia Anemia

### **Head to Toe Examination:**

Hair

Madarosis

Eye signs

Pretibial edema

Warm , Moist hands

Sweatiness

Shaky hands

Facies :

Hypothyroidism: Moon facies, Expression less , mask like , puffy

Hyperthyroidism: staring, excited, irritable, nervous , tense face

IQ (low)

Oral cavity (tremors , lingual thyroid)

Gait

Hyperthyroidism: slow lethargic

Hyperthyroidism; Rapid gait

Assessment of voice:

Pitch :

Ability to alter rapidity of speech

Breathing support during speech

Indirect Laryngoscopy

Position of vocal cord

Increased thickness

Sluggish movement

(say 'E' to demonstrate vocal cor movement)

One side medialisation – Hoarseness

Both sides medialisation – Stridor

Pallor, Icterus, Cyanosis, Clubbing, Koilonychia, Lymphadenopathy, Edema

### **Vitals:**

- Pulse:

Rate – Tachycardia in Toxicosis

Rhythm – regular/regularly/irregularly irregular ( ectopics)

Character Pulsus paradoxus or corrigans pulse

Volume – High volume in Toxicosis

Condition of vessel wall

Radio-radial/radio-femoral delay

- BP: (position and limb in which BP is measured) (HTN in Toxicosis)
- RR:
- Sleeping Pulse rate:  
Consent taken  
3 consecutive night readings  
Done in late nights towards early mornings
- Temperature:

|                                   |   |             |
|-----------------------------------|---|-------------|
| Crile's grading of sleeping pulse |   |             |
| Mild                              | - | 90-100 bpm  |
| Moderate                          | - | 100-110 bpm |
| Severe                            | - | >110 bpm    |

## **LOCAL EXAMINATION**

- Position of Patient: Patient was examined in sitting position with neck flexed slightly and patient examined from front
- Exposure:
- Position of Examiner wrt. Patient:
- Light source

## **Inspection:**

- Movement with deglutition
- Swelling:
  - Number
  - Situation: sides or centre in neck
  - Size
  - Shape
    - Butterfly – diffuse goitre
    - Hemispherical/globular in Solitary Nodular Goitre
  - Extent ( in diffuse swelling, describe 4 direction extent of each lobe and upper and lower border of isthmus)
  - Surface
    - Nodular – MNG
    - Irregular – Anaplastic Ca.
    - Smooth – SNT
  - Borders ( generally well defined)
  - Skin over swelling:
  - Scars ( previous surgeries)
  - Pigmentation
  - Venous prominence over gland and neck
  - Pulsatile – thyrotoxicosis
  - Movement with protrusion of tongue
- **Trail's sign**- prominence of SCM on side of deviation of trachea
- **Pemberton sign** ( if lower border not visible)
  - Raise both arm over head, until they touch the ears
  - Maintain the position for a while
  - Congestion of face and distress occurs due to obstruction of great veins of thorax

## **Palpation:**

Patient sitting comfortably. palpate with neck partially flexed , examiner being behind the patient. Examine with thumb placed on cervical spine and palpate with remaining fingers anteriorly to feel the gland both lateral lobes and isthmus for all features.. Additional information about lobe is obtained by relaxing SCM of that side by flexing and rotating face to same side.

## **Other methods**

**Crile's method** of palpation of gland: It is the palpation of the nodule/swelling in front using the pulp of the thumb.

## **Lahey's method**

To palpate right lobe push the right lobe with right hand to right side and palpate with left hand

## **Pizzilo's method**

In case of obese and short necked individuals

Ask the patient to keep the hands behind the head and ask to push head backwards against clasped hand on occiput

- Palpate for Isthmus in midline against Trachea ( normally not palpable)
- Temperature over swelling: ( warm in toxic, malignancy, thyroiditis)
- Tenderness ( tumour necrosis, Haemorrhage, Thyroiditis)

## **Swelling:**

- Extent, position & shape
- Palpate the lower border:
- Size vertical and horizontal measurement
- Surface :
  - Smooth - Colloid goitre, Grave's disease
  - Bosselated - Multinodular goitre
- Consistency :
  - Soft - Colloid goitre, Graves disease
  - Firm - SNG, MNG
  - Hard - Carcinoma, Riedel's thyroiditis
- Margins
- Fixity to skin
- Mobility : Restricted in malignancy and chronic thyroiditis
- Plane of swelling
  - \*Stretching of neck: deep fascia stretches and swelling becomes less prominent
  - \*Chin Compression against resistance: both SCM become prominent
  - \*SCM contraction- turn neck to one side against resistance
  - \*Strap muscles : ask patient to hold tongue against palate and deglute

- Palpate for thrill
- Measurement of circumference of the neck at regular intervals is important if patient is not undergoing surgery to assess the increase in size of the thyroid.
- **Position of trachea and larynx**
- **Pressure effects:**
  - Trachea - **Kocher's test**  
Patient is asked to see straight. With fingers and thumb both lateral lobes of the thyroid gland are gently compressed directing posteromedially. If patient develops stridor-Kocher's test is positive. If patient develops no stridor, it means test is negative

In long standing cases Trachea is kept patent because of forward traction by goitre itself. But after thyroidectomy lack of support to trachea cause tracheomalacia— weakening of the tracheal rings. Such patients need tracheostomy after thyroidectomy. It is usually temporary tracheostomy for 2-3 weeks, by then tracheal rings regain their strength to maintain the patency of the trachea. '**Scabbard trachea**' is narrowing of trachea

- Carotid artery - Carotid sheath is pushed back by benign swelling where carotid pulsations felt.\

Normally carotid pulse is felt at the level of the upper border of thyroid cartilage over medial aspect of the sternomastoid muscle on the Chaissagne tubercle (carotid tubercle) on the transverse process of C6 vertebra. It may be deviated posteriorly/laterally in a large goitre. It may be absent in advanced carcinoma thyroid due to infiltration of the carotid sheath by the tumour (Berry's sign).

- Sympathetic trunk - Horner's syndrome
  - Enophthalmos
  - Miosis
  - Anhidrosis
  - Ptosis

### **PERCUSSION:**

Over manubrium to Rule out Retrosternal extension.

Moses sign: pain on direct percussion over sternum in leukaemia

### **AUSCULTATION:**

Any audible bruits near upper pole

### **Examination of Cervical Lymph Nodes: (level III, IV, V & VI)**

It is common in papillary carcinoma of thyroid. It is usually in level III & IV nodes. It could be firm, hard or cystic. It is usually brownish black in colour often with papillary projections. Lymph nodes often can get enlarged in follicular carcinoma thyroid and lymphoma.

### **Examination for Primary Thyrotoxicosis:**

- Pulse
- Tachycardia
- Tremors
- Warm Moist skin
- Thyroid bruit
- Eye Signs
  - a. **Lid retraction:** Overactivity of sympathetic part of levator palpebrae superioris (Muller's muscle)
  - b. **Exophthalmos.**
    - i. Naffziger's test: Go behind the patient, extend the neck, see through the supraciliary ridge, you can diagnose exophthalmos.
    - ii. Gifford's test: Test to differentiate exophthalmos from proptosis (where pathology is behind the eyeball)
    - iii. Evert the upper eyelid. Impossible to do it in exophthalmos due to Muller's
    - iv. muscle hyperactivity. This test is possible in proptosis.
    - v. Stellwag's sign - Starring look with infrequent blinking and wide palpebral fissures.
    - vi. von Graefe's sign - Lid lag sign. Tested by asking the patient to look up and down many times fixing the head, you can see the upper lid lags behind.
    - vii. Joffroy's sign - Absence of wrinkling of forehead. The patient looks the roof of the room without forehead wrinkling.
    - viii. Dalrymple's - Visible upper sclera due to lid retraction.
    - ix. MÖbius sign - Inability to converge the eyeball.
    - x. Jellinek's sign - Increased pigmentation of eyelids.
  - c. **Ophthalmoplegia** (malignant exophthalmos)
    - Weakness of ocular muscles due to edema and cellular infiltration of these muscles.
    - Paralysis of superior rectus, inferior oblique and lateral rectus.
    - On paralysis of these muscles, patient is unable to look upwards and outwards.
  - d. Chemosis: conjunctiva edematous, thickened ,

**Examination for secondary Thyrotoxicosis:**

Complication of multinodular goiter and adenoma.

Signs of cardiac failure:

- Edema of ankles
- Orthopnoea
- Dyspnoea
- Eye signs

Only lid lag and lid retraction can be seen

- No tremor
- No exophthalmos

**SIGNS OF HYPOTHYROIDISM**

- Obese
- Dry inelastic skin
- Macroglossia
- Mask like facies
- Loss of hair in lateral eyebrow
- Hoarseness of voice
- Pseudomyotonic reflex (delayed ankle jerk); also called 'hung up' reflex.

Involvement of sympathetic chain: Horner's syndrome

**Systemic Examination:**

Cardiovascular system - Secondary thyrotoxicosis

RS - To find secondary deposits

Abdomen - In lymphoma of thyroid, other lymphoid organs in abdomen may be involved.

CNS: Myopathy Neuropathy

Musculoskeletal System: Spinal tenderness, Pulsatile metastasis to skull & ribs

Oral cavity - Look for lingual thyroid, macroglossia, tremor of tongue.

**Diagnosis:**

Anatomical diagnosis : MNG/SNG/Diffuse

Functional diagnosis : Toxic/Euthyroid/Hypothyroid

Pathological diagnosis : Benign/Malignant

## PERIPHERAL VASCULAR DISEASE

**Name:**

**Age:** TAO - <50 years  
Atherosclerosis - Old age  
Raynold's - Young age  
Diabetic arteriopathy - Middle age

**Sex:**

TAO – Common in Males  
Raynold's - Common in women

**Occupation:**

**Address:**

### CHIEF COMPLAINTS

- Pain
- Blackish Discolouration of toes / feet
- Ulcer

### HISTORY OF PRESENTING ILLNESS:

#### **1. Pain:**

- Site :** initially in calf later in foot( gangrene changes)
- Character :** Cramp like in calf and burning type (gangrene changes) in foot
- Intermittent Claudication:** distance of walking at which pain onsets ( previous and present distance is noted)
- Claudication distance:** Distance at which pain compels him to take rest from walking
- Radiation:**
- Aggravating factors:**
- Relieving factors:** (arterial disease pain do not relieve on medication)
- Presence of rest pain**—its location/severity/whether the pain gets relieved a little bit by holding the limb/foot/leg/toes (pain slightly lessens probably by transmission of temperature from holding hand into the part) or hanging the leg down or by applying the Warmth
- Progress of claudication:**
- Effect of heat and cold on pain**
  - TAO - warmth increase symptoms
  - Raynold's - symptoms increases on exposure to cold

#### **2. BLACKISH DISCOLOURATION:**

- History of trauma
- Site
- Onset
  - Atherosclerosis/Buerger's disease - spontaneous and gradual progression.*

- 2. *Embolism - sudden in onset, rapidly progressive with radiating severe pain along the artery.*
- 3. *Diabetic - from trauma*
- iv. Progression ( initially tip of toes & progressed to whole toe/foot)
- v. Extent of discolouration
- vi. Associated with pain??

### 3. Ulceration

- i. Whether precipitated by trauma/spontaneously.
  - ii. Pain in the ulcer/type/duration/aggravating or relieving factors.
  - iii. Discharge-type—serous-purulent-bloody.
  - iv. Progression.
4. History of **colour changes** in limb
  5. **Limitation of walking**—as the result of muscle pain is an important complaint.
  6. History of suggestive of **superficial thrombophlebitis** (30% TAO is associated)
    - i. Pain, swelling along course of superficial veins of lower limb
    - ii. Discolouration along course of superficial veins of lower limb
  7. History suggestive of **TIA**
    - i. Syncope, Blackouts
    - ii. Transient loss of consciousness, fainting
    - iii. Blurred vision
  8. History suggestive of **Mesenttric ischaemia**
    - i. Abdominal pain ( after having food)
    - ii. Bloody diarrhoea
  9. History suggestive of **Angina**
    - i. Exertional chest pain, Palpitations
  10. History of tingling/numbness/weakness in the limbs/pins and needles sensation in the skin of foot and leg **paraesthesia** due to shunting of cutaneous blood to deeper muscles
  11. History of **Impotence**: Its duration has to be asked [due to bilateral internal iliac artery (aortoiliac) block (Leriche syndrome)—present with pain in buttock].
  12. History of distal Neurological deficit
  13. History of haemoptysis, breathlessness ( **Pulmonary Embolism**)
  14. History of Haematuria ( **Renal Infarct**)
  15. History of repeated episodes or cyclic episodes of pain, paleness, bluish discoloration, dusk red colouration, associated with tingling, numbness, burning sensation on exposure to cold ( **Raynolds phenomenon**)
  16. History of edema pigmentation or ulceration around ankle (**venous diseases**)

### PAST HISTORY:

1. Similar history earlier.
2. History of drug intake earlier for similar conditions like vasodilators/drugs to increase the perfusion.
3. History of earlier surgery like sympathectomy/ omentoplasty/their results or effects.

4. History of exposure to cold
5. History of earlier cardiac illness( AF > embolism)
6. History of DM, HTN, Hypercholesterolemia, Collagen diseases

### **PERSONAL HISTORY:**

- **Appetite**
- **Diet**
- **Bowel and Bladder:**
- **Sleep**
- **Habits**

### **Family History:**

- History of Atherosclerosis
- History of Impotence

### **Treatment/Drug History:**

- Drug treatment to relieve pain
- Previous surgical treatment already done ( amputation)

### **Allergy History:**

### **General Physical Examination:**

Here is a \_\_\_\_ year old \_\_\_\_ patient, \_\_\_\_\_ built and nourished and \_\_\_\_ co-operative.

Pallor, Icterus, Cyanosis, Clubbing, Koilonychia, Lymphadenopathy, Edema

#### **Vitals:**

- Pulse: (rate, rhythm, character, volume, condition of vessel wall radio-radial/radio-femoral delay)
- BP: (position and limb in which BP is measured)( in all 4 limbs)
- RR:
- Temperature:

### **Head to toe Examination:**

- Signs of CCF
  - JVP
  - Pedal edema
  - Hepatomegaly
  - Ascitis
- Visual defects- loss of temporal field vision due to atherosclerosis of vessels

### **LOCAL EXAMINATION:**

- Position of Patient:  
Patient examined in lying position with outstretched legs
- Exposure:
- Position of Examiner wrt. Patient:
- Light source:

**INSPECTION:** ( keep both limbs side by side)

**1. Attitude of limb**

**2. Any bony deformity**

**3. Any muscle wasting** in thigh, calf or foot

**4. Condition of veins:**

- i. Whether normally filled veins in both limbs
- ii. Any discolouration along veins
- iii. Guttering of veins

**5. Signs of peripheral ischemia**

- i. Thinning of skin
- ii. Diminished hair
- iii. Lose of subcutaneous fat
- iv. shininess
- v. Trophic changes in nail: Brittle nails transverse ridges develop
- vi. Minor ulcerations in pressure areas  
( heel, malleoli, tip of toes, ball of foot 5<sup>th</sup> metatarsal head)
- vii. Gangrenous changes

**6. Colour of limb:**

- i. Minor changes in colour is noticed by comparing both legs
- ii. Major or marked pallor in sudden arterial obstruction like embolism or sudden spasm as in raynold's disease
- iii. Congestion & bluish cyanosed in ischaemia with pregangrenous features, develops pallor on limb elevation

**7. Gangrene**

- i. Site
- ii. Extent
- iii. Type of gangrene – **dry**(mummified) or **wet**(putrifying, oedematous)
- iv. Line of demarcation
  - 1. level & depth ( between gangrene tissue and viable tissue)
  - 2. well defined in dry gangrene
  - 3. ill defined and unclear in wet gangrene.
- v. Colour of gangrenous area –
  - 1. Black/purple/greenish black
  - 2. Reddish black [in gas gangrene (H<sub>2</sub>S)]
- vi. Discharge from area
- vii. Odour of discharge from gangrenous area .

**8. Observe limb above gangrenous area**

- i. Pale, congested, edematous
- ii. Black patches – skip lesions
- iii. Pre-gangrenous area ( ROHIT)
  - 1. Rest pain
  - 2. Oedema
  - 3. Hyperesthesia
  - 4. Ischaemic ulcer
  - 5. Temperature and colour changes

### **PALPATION:**

- **Skin temperature:** over affected limb ( warm / cold)
- Level of equalization of temperature:
  - Palpate with dorsum of hand from dorsum of foot upwards and find the level at which the temperature becomes normal comparing with normal area temperature of patients opposite limb.
- **Tenderness**
  - Site extend and severity should be noted
- **Palpation of gangrene and adjacent area:**

|                                | Gangrene | Pregangrenous area | Adjacent limb |
|--------------------------------|----------|--------------------|---------------|
| 1. Site and Extent             |          |                    |               |
| 2. Sensation                   |          |                    |               |
| 3. Tenderness                  |          |                    |               |
| 4. Local crepitations          |          |                    |               |
| 5. Edema (pitting/non pitting) |          |                    |               |

### Special Tests for Assessment of Circulatory Insufficiency

#### **1. BUERGER'S test:**

Test carried out in daylight, Patient lies supine, Raise the legs one after other with knees extended

**Normal limb:** Remain pink even when raised to 90°

Look for:

- \* Onset of pallor
- \* Guttering of veins

- Angle at which pallor appears is *Buerger's angle of circulatory insufficiency*
- Less than 30° implies severe ischemia

➤ If foot does not become pale or when doubtful, repeated ankle flexion and extension is done to point of fatigue.

- ❖ In occlusive arterial disease sole of foot becomes pale (cadaveric pallor) with empty-guttered veins on the dorsum of foot
- ❖ Cyanotic congestion appears after lowering the foot in 3 minutes.

#### **2. CAPILLARY REFILLING:**

##### **Inspection:**

Elevate limb for a while.

Ask the patient to sit and hang his legs down the bed

**Normal limb:** pink in elevated as well as in dependent position

**Ischaemic limb:** becomes pale on elevation & gradually becomes purple-red and then pink in more than 20 seconds. Purple-red colour is due to deoxygenated blood.

Prolonged capillary filling time signifies severe ischemia.

##### **Palpation:**

Tip of the nail or pulp of the finger or toe is pressed to blanch it and pressure is released (in 2 seconds) to make it pink again. Time taken for blanched area to turn pink is *capillary refilling time*.

It is prolonged in ischaemic limb

### 3. VENOUS REFILLING

#### Inspection:

After keeping the limb elevated for a while if it is laid flat on the bed, there will be normal refilling of veins within 5 seconds; but in ischemic limbs it will be delayed.

#### Palpation: Harvey's venous refilling test:

- Two index fingers are placed over the vein and pressure is applied over it. Proximal finger is moved for about 5 cm proximally without releasing the pressure.
- Vein between the fingers gets emptied completely and becomes flat.
- Distal finger is now released to see the flow of the blood and its refilling is observed,
- whether good or poor.
- It is poor in ischaemic limb.

### 4. Crossed leg test (Fuchsig's test):

- Indirect test for assessment of presence of popliteal pulse.
- Patient is asked to sit with the legs crossed one above the other so that the popliteal fossa of one leg will lie against the knee of other leg.
- Oscillatory movements of foot can be observed synchronous with the popliteal artery pulsation.
- If the popliteal artery is blocked oscillatory movements will be absent

### 5. COLD AND WARM WATER TEST

To provoke angiospasm in Raynold's phenomenon

Patient is asked to dip his hand into

Ice cold water - hand becomes white

Warm water - blue due to cyanotic congestion

### 6. MOVEMENTS OF JOINTS & LIMB ADJACENT TO GANGRENOUS AREA

- Movements of
- a. interphalangeal joints
  - b. mid tarsal joint
  - c. ankle joint
  - d. knee joint
  - e. hip joint

### 7. EXAMINATION OF NERVE LESIONS IN LOWER LIMB:

#### Sensory

- Crude touch/fine touch
- Pain sensation
- Temperature sensation
- Reflexes
- Plantar response

#### Motor

- Tone
- Power of muscles
- Ankle flexion/extension
- Knee flexion/extension

*“On the borderline of gangrene, skin is hyperesthetic”*

### 8. Examination of Regional lymph nodes ( Inguinal)

## 9. PALPATION OF PERIPHERAL PULSUS:

(++ normal , + palpable but feeble, - not palpable)

| Pulse                                                                                                                                                                                                            | Right | Left |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|
| 1. Dorsalis pedis<br>2. Posterior tibial<br>3. Anterior tibial<br>4. Popliteal<br>5. Femoral<br>6. Radial<br>7. Ulnar<br>8. Brachial<br>9. Axillary<br>10. Subclavian<br>11. Carotid<br>12. Superficial temporal |       |      |

- Condition of arterial wall ( Atherosclerosis )
- Tenderness
- Auscultation along major arteries for Bruit over major arteries.
  - ❖ Carotid: along anterior border of SCM at level of thyroid cartilage
  - ❖ Abdominal Aorta: Midline , 4 cm above transpyloric line extends to a point 2cm below and left to umbilicus ( upto L4)
  - ❖ Ileac: L4 to a point midway between ASIS and pubic symphysis
  - ❖ Coeliac: 4cm above transpyloric plane
  - ❖ Superior mesenteric: 2cm above transpyloric plane

### Palpation of various pulse:

1. **Dorsalis pedis artery** is felt just lateral to the extensor hallucis longus tendon at the proximal end of first web space, felt against the navicular and middle cuneiform bones. It is absent in 10% cases.
2. **Posterior tibial artery** is felt against the calcaneus just behind the medial malleolus midway between it and tendo-Achilles.
3. **Anterior tibial artery** is felt in the midway anteriorly between the two malleoli against the lower end of tibia just above the ankle joint lateral to extensor hallucis longus tendon.
4. **Popliteal artery**
  - Supine: Flexing knee 40°, heel over bed; with thumbs over tibial tuberosity; other fingers are moved sideways to palpate popliteal artery over the posterior aspect of tibial condyles.
  - Prone (knee flexed): Popliteal pulse felt over the posterior surface of lower end of femur.
  - Fuschig's test: Crossed legs on sitting, inspect the oscillatory movements sitting on chair
5. **Femoral artery** in the groin is felt just below the inguinal ligament midway between anterior superior iliac spine and pubic symphysis (midinguinal point). Often hip has to be flexed for about 10-15° to feel it properly.
6. **Radial artery** is felt at the wrist on the lateral aspect against lower end of the front of radius.
7. **Ulnar artery** is felt at the wrist on the medial end against lower end of the front of ulna.
8. **Brachial artery** is felt in front of the elbow just medial to biceps brachii tendon.
9. **Axillary artery** is felt in lateral aspect of the axilla against upper end of the shaft of the humerus with raised and elevated arm.
10. **Subclavian artery** is felt against first rib just above the middle of the clavicle in supraclavicular fossa while patient is lifting the shoulder to relax deep fascia.

11. **Common carotid artery** is felt medial to sternomastoid muscle at the level of thyroid cartilage against carotid tubercle (Chassaigne tubercle) of transverse process of 6th cervical vertebra (in carotid triangle).
12. **Facial artery** is felt against body of mandible at the insertion of masseter.
13. **Superficial temporal artery** is felt just in front of the tragus of the ear against zygomatic bone.

## **Diagnosis:**

Peripheral Arterial Occlusive Disease affecting Right / Left limb with/without complication of gangrene in the \_\_\_\_ (site) or ischaemic ulcer over \_\_\_\_ (site) due to obstruction at \_\_\_\_ level/segment probably due to

- Buerger's Disease
- Atherosclerotic Vascular Disease

## **Management:**

### **Investigations:**

#### ***Blood Investigations***

- Blood sugar and urine sugar
- Lipid profile (LDL, HDL, TG, total cholesterol)
- Anemia and conditions causing high blood viscosity (polycythemia and thrombocythemia)
- ESR
- Plasma fibrinogen
- Serum creatinine (hypertension)—to give contrast agents
- Coagulation profile: Prothombin time, bleeding and clotting time.

#### ***cardiovascular system assessment***

*X-ray Chest*

*ECG*

*Echocardiogram*

*Ankle-brachial pressure index (ABPI)*

$$\text{Right ABPI} = \frac{\text{Higher right Ankle pressure (DP/PT)}}{\text{Highest brachial artery pressure (R/L)}}$$

Normal : 0.9–1.2

#### ***Exercise Tests (Increases Ankle Pressure)***

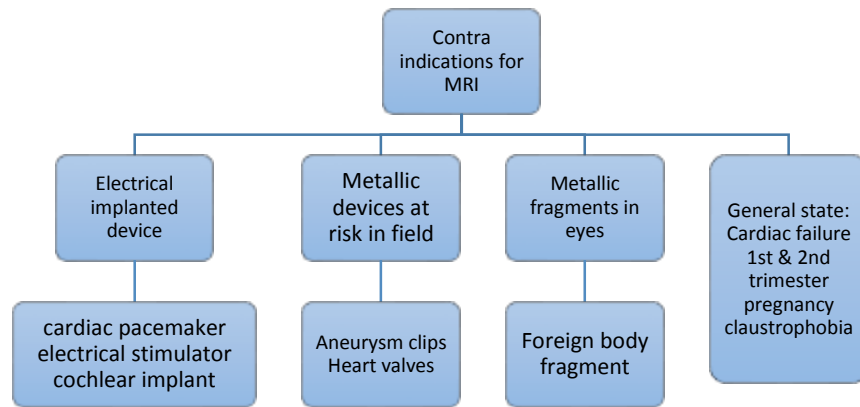
- 'To evaluate asymptomatic lesions'
- 'During exercise increased blood flow to muscle leads to fall in ankle pressure in the presence of arterial disease'
- Exercise protocol: Walk on treadmill 3.5 km/hr for 5 minutes ABPI is measured again, fall in ABPI is indicative of degree of ischemia.

***Angiography (Gold standard for lower limb assessment)***

***DSA ( Digital Subtraction Angiography)***

#### **Noninvasive Vascular Assessments**

- ***Hand held Doppler***
- ***Duplex scanCT angiogram***
- ***MRA ( Magnetic Resonant Angiogram)***



- **U/S Abdomen**
- **Plain X-Ray of the part**

#### **Transcutaneous oximetry**

### **Treatment Options for Peripheral Vascular Disease**

*Conservative*

*Medical*

*Surgical—minimally invasive and open.*

#### **Modify the Risk Factors**

- *Smoking: To be stopped*
- *Diabetes control: Maintain normal blood sugar*
- *Hypertension and heart disease corrected*
- *Correct hyperlipidemia: LDL < 100 mg/dL: By giving statins*
- *Obesity—weight reduction*
- *Hyperhomocysteinemia > 5 mmol/L must be considered for treatment Vitamin B<sub>12</sub> and folic acid*
- *Exercise programs: 1 hour/day; 1 to 3 times weekly for 6 months.*

#### **Medical:**

##### **Established drugs with proven but small benefit in improving claudication:**

• **Pentoxifylline** : Improves red cell deformability, lowers serum fibrinogen, reduces platelet aggregation  
No definite evidence to show it is better than placebo.

• **Naftidrofuryl** : 5-hydroxytryptamine antagonist  
Reduces platelet aggregation

• **Cilostazol** : Phosphodiesterase III inhibitor  
Antiplatelet and vasodilator activity

##### **Established drugs with minimal or no benefit in improving claudication:**

• **Aspirin** : Low dose aspirin still remains the mainstay of therapy Inhibits the cyclo-oxygenase  
Thereby decreases ADP release and platelet aggregation

• **Dipyridamole** : Along with aspirin  
Reduces stroke rate in atherosclerotic disease

• **Clopidogrel** : GP2b3a inhibitor  
Prevents platelet aggregation

#### **Surgery : ANGIOPLASTY**

# VARICOSE VEINS

Name:

Age:

Sex: more common in females 10:1

Address:

Occupation: More common in jobs having prolonged standing,  
e.g. policeman, petrol bunk workers, tram drivers, etc.

## Chief Complaints:

- Asymptomatic in 50% cases
- Pain
- Postural discomfort
- Swellings (Appearance of varicosity) in leg
- Ulcers
- Pigmentation

Varicosity is seen in:

- Lower limb
- Esophageal varices
- Hemorrhoids
- Varicocele testis
- Vulval varices
- Ovarian varices
- Gastric varices

## History of Presenting Illness

### **Pain:**

- Site : along the line of the vein is typical
- Onset :
- Duration:
- Character:
- dragging type in calf. Feeling of heaviness is common
- Bursting type of pain while walking (Deep vein thrombosis)
- Aggravating or Relieving factors: usually aggravates in the evening and gets relieved by lying down.
- Pain in calf of short duration, may be due to coexisting deep vein thrombosis

### **Swelling:** ( multiple swellings or varicosities }

- Site : along the line of the vein is typical
- Medial side of Lower limb: Large Saphenous System
- Lateral side of Lower limb: Small Saphenous System
- Onset :
- Duration:
- Aggravating or Relieving factors:
- History of swelling around ankle: (edema )(all routine history about swelling)
- duration;
- Diurnal variation:
- Postural variation: relation to work/standing/ lying down should be noted.

History suggestive of difficulty/ altered gait due to pain/swelling/deformity

### **Pigmentation**

- It is due to stasis and release of chemicals and usually occurs around ankle region.
- It is associated with itching and often ulceration.

### **Ulcer**

- Onset
- Duration
- Site
  - on the medial aspect of the ankle is due to long saphenous vein varicosity;

- on the lateral aspect is due to short saphenous vein varicosity.
- Discharge from ulcer—its type, smell, quantity signifies the severity of the infection.
- Itching and bleeding in the ulcer bed
- History of Trauma
- History of bleeding from the vein/ulcer
- History of urinary/bowel symptoms.
- Constipation
- Blood tinged stools
- Burning micturition
- Frequent micturition/ urgency
- History of abdominal distention, ascites
- History of lump or pain in abdomen  
(Abdominal mass/pregnancy may compress IVC/iliac veins and cause bilateral varicose veins)
- History of similar complaints on the other leg—Varicose veins are often bilateral.

#### **Past history:**

- History of prolonged immobilization ( Hospitalization)
- History of similar complaints
- History suggestive of earlier deep vein thrombosis like pain, calf swelling and fever should be noted.
- History of any previous surgery
- H/o DM, HTN, IHD, Asthma,

#### **Personal History:**

- Appetite:
- Diet:
- Bowel & Bladder:
- Sleep:
- Habits

Pregnancy history in females is important

- delivery
- postdelivery period
- oral contraceptive intake

#### **Family History**

- Similar history in the family. Inheritance with FOXC2 gene.

#### **Drug history:**

- OCPs
- Allergic History

## **GENERAL PHYSICAL EXAMINATION:**

Here is a \_\_\_\_ year old \_\_\_\_ patient, \_\_\_\_\_ built and nourished and \_\_\_\_ co-operative.

Pallor, Icterus, Cyanosis, Clubbing, Koilonychia, Lymphadenopathy, Edema

Vitals:

- Pulse: (rate, rhythm, character, volume, condition of vessel wall radio-radial/radio-femoral delay)
- BP: (position and limb in which BP is measured)
- RR:
- Temperature:

## **LOCAL EXAMINATION**

- Position of Patient: standing position, for some tests patient asked to lie down
- Exposure: below thigh
- Position of Examiner wrt. Patient:
- Light source: daylight is best

## **INSPECTION:**

- Dilated, elongated, tortuous veins on
  - Medial side : long saphenous vein
  - Posterior /lateral side : short saphenous vein.
- Swelling of the leg:
  - Localized ( calf / foot/ thigh) : superficial system is involved
  - Generalized : may be DVT
- Skin Changes:
  - Pigmentation
  - Edema
  - Ankle flare
  - Ulcer
    1. Site :
      - i. Medial malleolus : LSV involved
      - ii. Lateral malleolus : SSV involved
    2. Size & shape: Vertically oval varicose ulcer
    3. Extent:
    4. Floor
    5. Margin & edges
    6. Discharges
    7. surrounding area
  - Impending gangrene changes
    1. Scars
    2. Loss of hair
    3. Brittleness of nail
  - Scratch marks ( itching due to haemosiderin)
  - Deformity : Talipes equinovarus

▪ ***Morrissey's Cough Impulse test:***

The varicose veins are emptied by elevating the leg in lying down position. Then the patient is asked to cough. If there is saphenofemoral incompetence, expansile impulse is seen at saphenous opening - *saphena varix*.

**PALPATION:**

Examine the lower limb for:

- Temperature & Tenderness
- Thickening of skin
- Pitting edema
- Redness
- Lipodermatosclerosis
- Varicose venous ulcer: (describe ulcer)

**Special tests:**

***Morrissey's cough impulse:*** ( lying position)

- Limb is elevated to empty the veins and the limb is then put to bed and the patient is asked to cough forcible
- An expansile impulse is felt at the saphenofemoral junction in cases of saphenofemoral incompetence.

**Saphenous opening:**

- Just 4 cm below and lateral to the public tubercle
- Closed by cribriform fascia and forms the lower boundary of femoral canal
- Saphenofemoral junction is seen just above the saphenous opening

***Brodie Trendelenburg tests:***

Test 1. For saphenofemoral incompetence

Test 2. For perforator incompetence

**Procedure:**

Step 1 : Patient in recumbent position legs raised to empty the vein, may be hastened by milking the veins.

Step 2 : Tourniquet is applied below saphenofemoral (SF) junction (Thumb may be used to occlude the SF junction).

Test 1 : Pressure released at the SF junction.

- Varices fill very quickly from above
- Test 1 is positive, i.e. saphenofemoral incompetence is present.

Test 2 : Do not release the pressure for one minute

- Gradual filling of veins occur in the lower limb.
- Test 2 is positive, i.e. perforator incompetence is present.

***Oschner's Mahoner's Test ( Multiple Tourniquet Test)***

Step 1 : Patient in recumbent position, empty all the veins

Step 2 : Three tourniquets are applied:

1. Just below saphenofemoral junction
2. Just below mid-thigh
3. Just below knee
4. below-knee and above ankle level

Tourniquets are applied below each perforator. Ask the patient to stand.

### Inference:

- Appearance of veins between tourniquet 1 and 2 is seen in adductor canal perforator incompetence
- Appearance of veins between tourniquet 2 and 3 is seen in below knee perforator incompetence
- Appearance of veins between tourniquet 3 and 4 is seen in lower leg perforators incompetent.
- Appearance of veins below 4th tourniquet implies lower leg perforator incompetence.

### **Short Saphenous Venous Incompetence**

- Method 1
  - Apply a tourniquet at the upper thigh, thus blocking the GSV blood to flow into femoral vein and diverting most of the blood into the popliteal vein
  - The blood from the popliteal vein backflows into SSV if there is saphenopopliteal incompetence and the SSV becomes more prominent.
- Method 2
  - Identify the saphenopopliteal junction (usually in the popliteal fossa) and occlude it
  - Ask the patient to stand
  - Release the pressure at saphenopopliteal junction
  - Veins fill rapidly on the lateral side if there is saphenopopliteal incompetence.

### **Schwartz test:**

- Ask the patient to stand and keep thumb of one hand at the saphenous opening
- Tap with other hand along the course of long saphenous vein in the lower part of leg.

Inference: Impulse is felt in the thumb at saphenous opening. This test implies the valves along the GSV are incompetent

### **Pratt's test:**

To mark the position of weak perforators (blow outs).

Steps:

- Apply Esmarch elastic bandage from toes to groin to empty the superficial veins.
- Apply tourniquet at groin (below SF junction).
- With tourniquet in position remove bandage gradually from above below and simultaneously apply another elastic bandage from groin to toes in reverse direction.

Inference: At the position of weak perforators blow outs can be seen. Mark these blow outs with skin pencil.

### **Fegan's test:**

- Line of varicose vein is marked. On standing, the site where the perforators enter the deep fascia bulges and these points are also marked. ( also by Pratt's test)
- In lying down, button like depressions (crescentic gaps) in the deep fascia are felt at the marked out points which confirms the perforator site.

***Ian-Aird test:***

On standing, proximal segment of long saphenous vein is emptied with two index fingers. Pressure from proximal finger is released to see the rapid filling from above which confirms saphenofemoral incompetence.

**Tests for DVT*****Modified Perthes test:***

- To find deep vein thrombosis
- Important preliminary to do this test is that there should not be any perforator incompetence to do this test.

**Steps:**

- Tourniquet is applied below the saphenofemoral junction (no need to empty the veins before applying tourniquet).
- Ask the patient to walk with tourniquet.

**Observation:**

- Shrinking of varicose veins: Indicates that there is normal deep veins and perforators.

(Note: If there is perforator incompetence there will not be shrinking of veins, hence cannot be done in cases of perforator incompetence)

- More prominence of varicose veins associated with severe cramp like pain: Indicates there is deep vein thrombosis.

***Homan's sign:***

Forcible dorsiflexion of foot with knee extension causes pain in the calf.

***Moses sign:***

Squeezing the calf muscles from side-to-side results in severe pain at the calf.

**Note:** These tests are not to be done nowadays for the risk of dislodgement of thrombus resulting in pulmonary embolism.

**PERCUSSION:** 'Schwartz test'

**AUSCULTATION:** 'Morrisey's cough impulse test'.

**Examine:**

Regional nodes Vertical inguinal nodes and external iliac nodes (above and medial aspect of the inguinal ligament) are palpated

Other limb

Ankle Joint movements (plantar and dorsiflexion) are checked for any restriction.

Measure both limbs circumference above or below a fixed bony point and mention any swelling of affected limb.

Arterial pulsations ( associated arterial disease)

## **SYSTEMIC EXAMINATION:**

*Cardiovascular system:* murmurs ( tricuspid incompetence)

*RS:*

*Abdomen:*

- Palpable mass
- Pregnancy
- Fibroid
- Ovarian cyst
- Carcinoma cervix or rectum.

*CNS:* for focal neurological deficit

## **Diagnosis:**

- Primary (or) secondary varicose vein
- Involving great saphenous (or) short saphenous (or) both venous system
- With or without saphenofemoral incompetence (or) saphenopopliteal incompetence
- With or without perforator incompetence
- With clinical class ( CEAP grade 0-6)
- With or without complications
  1. Due to dilated veins:
    - Hemorrhage
    - Phlebitis (Thrombosis)
    - Calcification of veins.
  2. Due to ulcer:

Ulcer complicates varicose vein in about 5 percent of the people.

    - Marjolin's ulcer : Malignant change in long-standing venous ulcer.
    - Periostitis tibia : Occurs in long-standing ulcer on medial surface of tibia.
    - Equinus deformity : Walking on toe relieves pain, so he continues walking resulting in shortening of tendo-Achilles.
  3. Skin changes:
    - Eczema
    - Pigmentation (due to hemosiderin deposition) Due to high venous pressure red blood cell (RBC) is forced into the capillaries resulting in hemoglobin break down to form hemosiderin.
    - Lipodermatosclerosis.

## **MANAGEMENT:**

*Investigations:*

- Duplex scan
  - Standing
  - Lying
  - Valsalva maneuver
- Hand Held Doppler
- Plethysmography
- Venography
  - Ascending (DVT)
  - Descending ( Valve incompetence)

- AVP: Ambulatory venous Pressure
- RAJU's test ( for venous obstruction) Arm – Foot venous pressure study

## **Treatment:**

### **Conservative therapy**

#### ***Indications:***

- *Pregnancy*
- *Pelvic tumour*
- *Perthe's positive patient*
- *AV fistula*
  - Elastic compression crepe bandage ( 30-40 mm Hg)
  - Elevation of limb (3-4 times a day for 30 min)
  - Unna boots
  - Exercises
  - Pneumatic compression ( not used commonly)

#### ***Sclerotherapy:*** ( contraindicated in DVT)

- Catheter Sclerotherapy
- Liquid Sclerotherapy
  - Micro sclerotherapy
  - Macro sclerotherapy
  - Trans illumination sclerotherapy
- Foam Sclerotherapy ( sclerosant + CO<sub>2</sub> + O<sub>2</sub>)
- Echo Sclerotherapy

#### **Sclerosants used:**

##### **Osmotic**

Chromated glycerine  
Hypotonic saline

##### **Detergent**

10ml of 3% STDS  
Polidocanol  
Sodium marrhavate  
Ethanolamine oleate

#### ***Complications of sclerotherapy***

- Allergy
- Pigmentation ( STDS)
- DVT
- Thrombophlebitis
- Skin necrosis

#### ***Surgery:***

- Trendelenburg operation and stripping
- Cockett and Dodd subfascial ligation
- *Linton's surgery ( don't mention)*
- *Hook Phlebectomy ( don't mention)*

#### ***Minimally Invasive Procedures***

- *SEPS – Subfascial Endoscopic Perforator ligation Surgery*

- *RFA Radio Frequency Ablation*
- *EVLA Endo Venous Laser Ablation*
- *TRIVex*
- *CHIEVA ( don 't mention)*

# HERNIA

**Name:**

**Age:** *\*Indirect Inguinal Hernia is common in Young*

*\*Direct Inguinal Hernia is common in elderly*

*\*Direct Inguinal Hernia never occurs in children and infants because superficial and deep rings are opposed to each other and no posterior wall exists*

**Sex:**

- *Most common hernia in females – Indirect Inguinal Hernia ( other than incisional)*
- *Most common henia in males – Indirect Hernia*
- *Femoral hernia is most common among females*
- *Direct Hernia never occurs in Females*

**Occupation :** *most common in strenuous workers*

**Address:**

**DOA & DOE:**

**Chief Complaints:**

- Swelling in the groin & Scrotum
- Pain over swelling
- Symptoms of Intestinal obstruction:
  - Colicky pain
  - Abdominal distention
  - Obstipation
  - Vomiting

**History of Presenting Illness:**

*I. About the hernia*

*a. **Swelling:***

- i. Duration
- ii. Onset: Suddenly/gradually
- iii. Site of start:
  - a. *From groin to scrotum (hernia)*
  - b. *From scrotum to groin (hydrocele and varicocele)*
- iv. Progression: *Size and extent of hernia at onset and at current stage*
- v. Aggravating factors:
  - a. *On straining*
  - b. *On standing*
  - c. *On coughing*
  - d. *If it appears Spontaneously – Direct Hernia*
    1. *Gradually - Indirect Hernia*
- vi. Relieving factors:
  - a. *By lying down*
  - b. *Manually by himself*
  - c. *If it disappears*
    1. *Immediately – Direct Hernia*
    2. *Gradually – Indirect Hernia*
    3. *Patient assisted – Indirect Hernia*
- vii. *Any period of irreducibility ( strangulated henia , obstructed hernia)*
- viii. *Any swelling on opposite side*

**b. Pain:**

- i. Site :
- ii. Onset:
- iii. Character :
  - a. Dull aching
  - b. Dragging
  - c. Vague discomfort
  - d. Colicky pain – obstruction
- iv. Radiation
- v. Duration and severity
- vi. Aggravating factors ( on straining)
- vii. Relieving factors (on lying down)

**II. Precipitating factors ( history of straining)**

- a) History of chronic cough and breathlessness
- b) h/o **Bowel obstruction**
  - i. *constipation*
  - ii. *straining at stools*
- c) h/o bladder disturbances
  - i. *dysuria*
  - ii. *hesitancy: person feels bladders is full & wants to pass urine but has to wait long to urine stream starts to flow*
  - iii. *urgency*
  - iv. *precipitancy: need to get to toilet in a hurry to prevent leakage*
  - v. *Frequency of micturition*
  - vi. *anuresis*

**III. Due to hernia (Complications)**

**History suggestive of Strangulation:**

*(Obstruction + irreducibility + Arrest of blood supply)*

*Colicky abdominal pain if continues and becomes gangrenous pain disappears*

*Sudden increase in size of hernia; becomes tense and tender.*

**History suggestive of Intestinal Obstruction**

*Colicky pain*

*Abdominal distention*

*Obstipation*

*Vomiting*

**PAST HISTORY:**

History of similar complaints in past on same side / opposite side

History of previous surgeries

*(injury to ilioinguinal nerve > Direct Hernia)*

*Appendicectomy*

*Palomo's operation for varicocele*

*Placing drain after surgery*

History of DM, HTN, Asthma, TB, IHD

## PERSONAL HISTORY

- **Occupation**
- **Appetite**
- **Diet**
- **Bowel and Bladder:**
- **Sleep**
- **Habits**
  - Smoking** – *chronic bronchitis*
  - *collagen deficiency*

### Family History:

History of connective tissue disorders in family.

### Drug/Treatment History:

History of use of TRUSS

### Allergy History:

### General Physical Examination:

Here is a \_\_\_\_ year old \_\_\_\_ patient, \_\_\_\_\_ built and nourished and \_\_\_\_ co-operative.

Pallor, Icterus, Cyanosis, Clubbing, Koilonychia, Lymphadenopathy, Edema

#### **Vitals:**

- Pulse: (rate, rhythm, character, volume, condition of vessel wall  
radio-radial/radio-femoral delay)
- BP: (position and limb in which BP is measured)( in all 4 limbs)
- RR:
- Temperature:

### LOCAL EXAMINATION:

- Position of Patient:  
Patient examined in standing position and in lying down position for some tests
- Exposure: umbilicus to mid thigh
- Position of Examiner wrt. Patient:
- Light source:

#### Rules of hernia examination

- Never forget to check expansile impulse on coughing and reducibility.
- Never forget to examine opposite side.
- Never forget to do per-rectal examination.
- Never forget to examine bulbar urethra.
- Never forget to check abdominal muscle tone.

## Inspection

1. Side
2. Site
3. Position
  - a. Femoral – below & lateral to pubic tubercle
  - b. Inguinal – above & medial to pubic tubercle
4. Size ( horizontal and vertical dimensions)
5. Extent:
  - a. Swelling in inguinal/inguino scrotal region
  - b. Swelling extend upto inguinal canal, below upto \_\_ cm above root of scrotum/upto bottom of scrotum. Medially \_\_ cm from midline to laterally \_\_ cm from ASIS.
6. Shape:
  - a. Pyriform – Indirect
  - b. Globular – Direct
  - c. Retort – Femoral
7. Surface
8. Margins
9. Skin over swelling
  - a. Scar
  - b. Engorged veins
  - c. Redness
  - d. Pigmentation
10. Reducibility of the content on lying down
11. Expansile cough impulse
  - No need to reduce the content
  - Just ask the patient to stand and turn face away and cough
  - Observe carefully at superficial inguinal ring
- Inference
  - Swelling increases in size if already exists
  - swelling reappears, synchronous with cough
12. Any visible peristalsis over swelling
13. Opposite scrotum inspection
14. whether testis is seen separately from the swelling or covered by the swelling all over
15. Position of Penis
16. Urethral meatus
17. Opposite scrotum

## Palpation:

1. Temperature
2. Tenderness
3. Site
4. Size
5. Shape
6. Extent
7. Surface
8. Skin over

|                                                                   |
|-------------------------------------------------------------------|
| Consistency:<br>Soft elastic—intestine<br>Doughy granular—omentum |
|-------------------------------------------------------------------|

9. Consistency

10. Reducibility

11. Get above the swelling (Get above the swelling is a classical feature of hydrocele)

12. Cough impulse

Propulsive and Expansile Impulse on Coughing'

Can be performed by:

- Making the child cry
- Valsalva maneuver
- Head raising and abdomen contraction

13. Ring Invagination test

Only test in hernia; done in lying position.

Prerequisite:

- Swelling should be reducible
- Lax of skin should be there for invaginating (so this test could not be done in females)

#### Interpretation of Invagination test

- Strength of superficial ring: Normal ring admits only the tip
- Direction of canal:
  - Direct hernia—directly backwards
  - Indirect—goes upwards, backwards and laterally
- Site of impulse:
  - Pulp—direct
  - Tip—indirect
- Strength of posterior wall
- To find early cases of hernia, impulse felt at tip

14. Ring occlusion test

After reducing the contents, patient in standing position, occlude the deep ring with thumb. Ask the patient to cough.

If swelling appears – Direct

Does not appear - Indirect

15. Zieman's technique :

For right side inguinal hernia, place the right hand

- Index finger over deep ring
- Middle finger over superficial ring
- Ring finger over saphenous opening

See where the impulse is felt

- Direct hernia—superficial ring
- Indirect hernia—deep ring
- Femoral hernia—saphenous opening

16. Palpation of testis, epididymis and spermatic cord should be done without fail.

Relation of swelling to testis also should be noted.

17. Bulbar urethra is palpated by lifting the scrotum and feeling in the midline. (To look for thickening and button like depression , a feature of stricture urethra).

18. Opposite inguinal region, opposite testis, epididymis and spermatic cord should be examined. Presence or absence of impulse on coughing on opposite side should be mentioned.

How will you demonstrate hernia in children?

### **Gornall's Test**

- Child held from back by both hands of clinician on its abdomen
- Abdomen is pressed and child is lifted up
- Hernia appears due to increase in the abdominal pressure exerted.

### **Percussion**

Without reducing contents of the swelling, percussion is done over the surface. If it is resonant, it is enterocele. If it is dull on percussion, then it is omentocele.

#### **TAXIS (Do not mention unless asked by the examiner)**

Method of reducing the inguinal hernia

Procedure: Flex the knee, Adduct and internally rotate the hip



Relaxes the abdominal muscles

\*With the thumb and fingers hold the sac; guide with other hand at superficial ring\*

#### **Complications of Taxis**

- ✓ Bowel injury
- ✓ Reduction en masse: Reducing the sac with the constriction being present at the neck; thereby making the hernia with obstruction to go into the abdomen
- ✓ Sac may rupture at its neck and the contents may be reduced extra peritoneally.

## ***Auscultation***

*Peristaltic sounds occasionally heard.*

### **Others**

1. Testis: 'Traction Test' to find whether the inguinal swelling is an Encysted Hydrocele of Cord.
2. Epididymis.
3. Penis:
  - Phimosis
  - Penile strictures
  - Pinhole meatus
4. Regional nodes.
5. Opposite groin.

### **Per-rectal Examination**

To Rule out:

1. Benign Prostate hypertrophy—micturition difficulty
2. Malignant obstruction
3. Chronic fissure—constipation

### **Respiratory system Examination:**

#### **Diagnosis**

- Side—right/left
- Type—indirect/direct
- Inguinal—femoral
- Complete/Incomplete
- Complicated/Uncomplicated
- Content—enterocele/omentocele

## DIFFERENTIAL DIAGNOSIS

| <i>Inguinal Swelling</i> | <i>Inguinoscrotal Swelling</i>     | <i>Femoral Hernia</i> |
|--------------------------|------------------------------------|-----------------------|
| 1. Enlarged lymph nodes  | 1. Encysted hydrocele of cord      | 1. Inguinal hernia    |
| 2. Undescended testis    | 2. Varicocele                      | 2. Saphenavarix       |
| 3. Lipoma                | 3. Lymphvarix                      | 3. Cloquet's node     |
| 4. Femoral hernia        | 4. Diffuse lipoma of cord          | 4. Lipoma             |
| 5. Saphena varix         | 5. Inflammatory thickening of cord | 5. Femoral aneurysm   |
| 6. Psoas abscess         |                                    | 6. Psoas abscess      |
| 7. Femoral aneurysm      |                                    |                       |

### Investegations

#### I. Routine

- Hemoglobin
- Bleeding time/Clotting time
- Total count, differential count, ESR
- Urine—albumin, sugar deposits
- Blood—urea, sugar
- Blood grouping/typing—for irreducible hernia/huge hernia

#### II. Anesthetic Purpose

- X-ray chest (Chronic TB, Asthma—precipitate hernia)
- ECG all leads

#### III. USG Abdomen and Pelvis

- In old age group—to find benign prostate hyperplasia calculate post voidal residual urine. If >100 ml it is significant
- To find any mass

Expansile impulse on coughing

- Hernia
- Meningocele
- Laryngocele
- Empyema necessitans
- Intracranially extended dermoid

#### Contents of spermatic cord

- Arteries :     Testicular Artery
  - Artery of Vas
  - Artery to Cremaster
- Veins :       Pampiniform plexus of veins
  - Veins corresponding to Arteries
- Lymphatics of testis
- Testicular plexus of sympathetic nerves
- Genital branch of genitofemoral N
- Vas deferens

### Mechanisms that prevent hernia when abdominal pressure rises.

1. Shutter mechanism—arched fibers of internal oblique
2. Flap valve mechanism—oblique canal; approximation of anterior and posterior wall.
3. Ball valve mechanism—cremaster contracts, thereby superficial ring plugged by spermatic cord.
4. Slit valve mechanism—crura of the superficial ring.

| <i>Indirect inguinal hernia</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <i>Direct inguinal hernia</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Can occur from childhood to adult.</p> <p>Occurs in a pre-existing sac.</p> <p>Protrusion through the deep ring.</p> <p>Herniation occurs later.</p> <p>Pyriform/oval in shape.</p> <p>Descends obliquely and downwards.</p> <p>Can become complete by descending down into the scrotum.</p> <p>Neck of the sac is narrow and lateral to inferior epigastric artery.</p> <p>Sac is anterolateral to the cord.</p> <p>Ring occlusion test does not show any impulse after occluding the deep ring.</p> <p>Invagination test shows impulse on the tip of the little finger.</p> <p>Zieman's test shows impulse on the index finger.</p> <p>Commonly unilateral but can be bilateral.</p> <p>Obstruction/strangulation are common.</p> <p>Sac should be opened during surgery.</p> | <p>Common in elderly.</p> <p>Always acquired.</p> <p>Herniation through posterior wall of the inguinal canal.</p> <p>Globular/round in shape.</p> <p>Descends directly forwards as a bulge.</p> <p>Descends down into the scrotum is rare.</p> <p>Neck of the sac is wide and medial to inferior to epigastric artery.</p> <p>Sac is posterior to the cord.</p> <p>Test shows impulse even after occluding the deep ring.</p> <p>Impulse is felt over the pulp of the little finger.</p> <p>Test shows impulse on the middle finger.</p> <p>Commonly bilateral.</p> <p>Rare but can occur.</p> <p>Sac is not necessarily opened unless—obstruction is present.</p> |

| <i>In enterocele</i>                                                                                                                                                                                     | <i>In omentocoele (epiplocele)</i>                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>First part is difficult to reduce but last part is easier There will be gurgling sound on reduction.</p> <p>Resonant on percussion.</p> <p>Peristalsis is seen.</p> <p>Bowel sounds may be heard.</p> | <p>First part is easier to reduce but last part is difficult. Has a doughy feeling.</p> <p>Dull on percussion.</p> <p>No peristalsis seen.</p> <p>Bowel sounds not heard.</p> |